Dr. Meghana Killi

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Education

Cosmic DAWN Center (DAWN), University of Copenhagen (UCPH) — Copenhagen, Denmark
M.Sc. & Ph.D. in Astrophysics, Supervisor: Darach Watson — 2018–2023

University of Texas at Austin (UT Austin) — Austin, USA
B.S.(Highest Honors) in Astronomy, Supervisors: Dr. Caitlin Casey and Dr. Volker Bromm — 2016–2018

Indian Institute of Technology at Kharagpur (IIT KGP) — Kharagpur, India
B.Tech.(Honours) in Mechanical Engineering — 2011–2015

Research Positions

Postdoctoral Researcher, **Centro de Astrofísica y Tecnologías Afines (CATA)** — Santiago & Concepción, Chile
Host: Diego Portales University — 2023–2024

Research Associate, **European Southern Observatory (ESO)** — Garching, Germany
Supervisors: Dr. Michele Ginolfi, Dr. Gergö Popping — Feb–Apr 2022

Undergraduate Research Assistant, **UT Austin** — Austin, USA
Supervisor: Dr. Caitlin Casey — 2016–2018

Teaching

Certifications and coursework
- PhD course in Introduction to University Pedagogy, UCPH — 2021
- PhD course in Advanced Presentation Techniques, UCPH — 2021
- Masters' course in Science Communication in Theory and Practice, UCPH — 2019

Teaching assistant
- Undergraduate lab in Quantum Mechanics II with over 100 students, UCPH — 2022, 2023
- Masters' course in Theoretical Astrophysics, including weekly tutorial sessions for over 30 students, UCPH — 2020

One-on-one tutor
- Undergraduate Calculus, Physics, and Astronomy at Texas Athletics Student Services, UT Austin — 2016
- Grade 10 Mathematics at the National Service Scheme (NSS), IIT KGP — 2011-2012

Scientific Talks & Posters

Astronomy Colloquium, University of Concepción — Concepción, Chile
"Evolution of teenage galaxies and active galactic nuclei (AGN)" — Apr 2024

CATA Scientific Meeting — Santiago, Chile
"Galaxy and AGN evolution through multi-wavelength observations" — Jan 2024

Monday Science Seminar Series, University of Wisconsin-Madison — Madison, USA (virtual)
"High-z galaxies: What is normal? What is not?" — Oct 2022

Charting the Metallicity Evolution History of the Universe — Catania, Italy
"A solar metallicity galaxy at $z > 7$" — Sep 2022

European Astronomical Society annual meeting — Valencia, Spain
"Spatial offsets between stellar and interstellar medium (ISM) phases in distant main-sequence galaxies" — Jun 2022
"[O III] 52µm and [N II] 122µm at $z > 7$; Novel metallicity constraint in the reionization epoch"

DAWN Summit — Copenhagen, Denmark
"Spatial offsets between stellar and ISM phases in distant main-sequence galaxies" — Jun 2022
"Compact Lyman-alpha emitters (LAEs) at high-z" — Sep 2021
"Lyman-alpha emitter morphology" — Jul 2020

Undergraduate Research Forum — Austin, USA, Apr 2017
Fall Undergraduate Research Symposium — Austin, USA, Sep 2016
Texas Astronomy Undergraduate Research Symposium — Waco, USA, Sep 2016
"Lessons on obscured star formation from ALMA archival data in COSMOS"

Scholarships & Awards

CATA Postdoctoral Research Fellowship for independent research, Chile 2023–Present

DAWN PhD Fellowship for the integrated M.Sc.+Ph.D. program, UCPH 2018–2023

General International Student and Scholar Services Financial Aid Award, UT Austin 2017

Best Presentation, Chemistry and Astronomy, UT Austin Fall Undergraduate Research Symposium 2016

John W. Cox Endowment for the Advanced Studies in Astronomy, UT Austin 2016

Dr. Ambedkar National Merit Scholarship, India 2009, 2011

Certificate of Merit from Narayana Group of Educational Institutions for academic excellence 2008–2009

Ranked 10th in the state of Andhra Pradesh, India in the Jr.APAMT (Andhra Pradesh Association of Mathematics Teachers) exam 2008

Technical skills

Programming: (proficient) Python

Astronomy data analysis: Morphology and spectroscopy including spectral cube collapse, source extraction, aperture photometry and flux extraction, centroid fitting, emission line identification and fitting, 1D line profile fitting, and plotting spectra and images

Telescopes and instruments: HST, JWST, VLT (KMOS), ALMA

Data reduction/Pipelines: ALMA, VLT (KMOS incl. P2UI, KARMA, EsoReflex)

Astronomy Programs: (proficient) CASA, GALFIT, msaexp, (basic) CARTA, EAZY, Bagpipes, TOPCAT, DS9, QFitsView, IDL

Other: (proficient) LaTeX, MSOffice, MacOS, ChromeOS (basic) GitHub, HTML, CSS, C++, JavaScript, SQL

Languages: (proficient) English, Telugu, Hindi, (basic) Korean, Spanish

Approved Telescope Proposals

Understanding Little Red Dots with ALMA and MIRI
P.I.: **Dr. Meghana Killi**; Project codes: 2024.1.01091.S (ALMA), Cycle 3 general observer program 6743 (JWST)
17.4 hours with ALMA and 13.7 hours with JWST/MIRI for a first-of-its-kind attempt to map the complete mid- to far-infrared spectra of five little red dot (LRD) candidates.

Dust in Galaxies at z=8–11
P.I.: Dr. Seiji Fujimoto; Project code: 2022.1.01562.S 2022
20 hours to observe ALMA Band 7 dust continuum of 23 bright Lyman Break Galaxy (LBG) candidates at z$\sim$8–11 with approved JWST Cycle 1 spectroscopy

The First Measurement of Metallicity and ISM Conditions of a Normal Galaxy at Reionisation
P.I.: Dr. Darach Watson; Project code: 2019.1.01778.S 2019
10 hours to measure [N II] and [O III] 52µm fluxes for A1689-zD1 in ALMA Bands 7 and 9 for the first metallicity estimate in the reionization epoch

Testing Long GRB SNe as a Source of r-process Material in the Universe
P.I.: Dr. Jonatan Selsing; DDT code: 2102.D-5025 2018
6 hours to observe the evolution of GRB171205A/SN2017iuk, $\sim$1 year after explosion using HAWK-I and X-SHOOTER instruments.

List of Publications

[1] A. Posses, ..., M. Killi, et al. "The ALMA-CRISTAL survey: Extended [CII] emission in an interacting galaxy system at z $\sim$ 5.5". In: A&A 699, A256 (July 2025), A256. DOI: 10.1051/0004-6361/202449843. arXiv: 2403.03379 [astro-ph.GA].

[2] R. Herrera-Camus, ..., M. Killi, et al. "The ALMA-CRISTAL survey: Gas, dust, and stars in star-forming galaxies when the Universe was $\sim$1 Gyr old: I. Survey overview and case studies". In: A&A 699, A80 (July 2025), A80. DOI: 10.1051/0004-6361/202553896. arXiv: 2505.06340 [astro-ph.GA].

[3] Belén Alcalde Pampliega, ..., and Meghana Killi. "Hiding behind a curtain of dust: Gas and dust properties of an ultra-luminous strongly-lensed z = 3.75 galaxy behind the Milky Way disk". In: *arXiv e-prints*, arXiv:2506.21283 (June 2025), arXiv:2506.21283. DOI: 10.48550/arXiv.2506.21283. arXiv: 2506.21283 [astro-ph.GA].

[4] N. E. P. Lines, ..., M. Killi, et al. "JWST PRIMER: a lack of outshining in four normal z = 4 - 6 galaxies from the ALMA-CRISTAL Survey". In: MNRAS 539.3 (May 2025), pp. 2685–2706. DOI: 10.1093/mnras/staf627. arXiv: 2409.10963 [astro-ph.GA].

[5] Ryota Ikeda, ..., Meghana Killi, et al. "The ALMA-CRISTAL Survey: Spatial extent of [CII] line emission in star-forming galaxies at z = 4-6". In: A&A 693, A237 (Jan. 2025), A237. DOI: 10.1051/0004-6361/202451811. arXiv: 2408.03374 [astro-ph.GA].

[6] K. E. Heintz, ..., M. Killi, et al. "The JWST-PRIMAL archival survey: A JWST/NIRSpec reference sample for the physical properties and Lyman-α absorption and emission of ∼600 galaxies at z = 5.0 - 13.4". In: A&A 693, A60 (Jan. 2025), A60. DOI: 10.1051/0004-6361/202450243. arXiv: 2404.02211 [astro-ph.GA].

[7] Juno Li, ..., Meghana Killi, et al. "The ALMA-CRISTAL Survey: Spatially Resolved Star Formation Activity and Dust Content in 4 ¡ z ¡ 6 Star-forming Galaxies". In: ApJ 976.1, 70 (Nov. 2024), p. 70. DOI: 10.3847/1538-4357/ad7fee. arXiv: 2409.10961 [astro-ph.GA].

[8] V. Villanueva, ..., **M. Killi**, et al. "The ALMA-CRISTAL survey: Dust temperature and physical conditions of the interstellar medium in a typical galaxy at z=5.66". In: *arXiv e-prints*, arXiv:2407.09681 (July 2024), arXiv:2407.09681. DOI: 10.48550/arXiv.2407.09681. arXiv: 2407.09681 [astro-ph.GA].

[9] **Meghana Killi**, Michele Ginolfi, Gergö Popping, et al. "The ALPINE-ALMA [C II] survey: characterization of spatial offsets in main-sequence galaxies at z 4-6". In: MNRAS 531.3 (July 2024), pp. 3222–3241. DOI: 10.1093/mnras/stae1371. arXiv: 2402.07982 [astro-ph.GA].

[10] Kasper E. Heintz, ..., **Meghana Killi**, et al. "Strong damped Lyman-α absorption in young star-forming galaxies at redshifts 9 to 11". In: *Science* 384.6698 (May 2024), pp. 890–894. DOI: 10.1126/science.adj0343. arXiv: 2306.00647 [astro-ph.GA].

[11] Abdurro'uf, ..., **M. Killi**, et al. "JWST NIRSpec High-resolution Spectroscopy of MACS0647-JD at z=10.167: Resolved [OII] Doublet and Electron Density in an Early Galaxy". In: *arXiv e-prints*, arXiv:2404.16201 (Apr. 2024), arXiv:2404.16201. DOI: 10.48550/arXiv.2404.16201. arXiv: 2404.16201 [astro-ph.GA].

[12] **M. Killi**, D. Watson, G. Brammer, et al. "Deciphering the JWST spectrum of a 'little red dot' at $z \sim 4.53$: An obscured AGN and its star-forming host". In: *arXiv e-prints*, arXiv:2312.03065 (Dec. 2023), arXiv:2312.03065. DOI: 10.48550/arXiv.2312.03065. arXiv: 2312.03065 [astro-ph.GA].

[13] K. E. Heintz, ..., **M. Killi**, et al. "Gauging the mass of metals in the gas phase of galaxies from the Local Universe to the Epoch of Reionization". In: *Astronomy & Astrophysics* 678, A30 (Oct. 2023), A30. DOI: 10.1051/0004-6361/202346573. arXiv: 2308.14813 [astro-ph.GA].

[14] **M. Killi**, D. Watson, S. Fujimoto, et al. "A solar metallicity galaxy at $z > 7$? Possible detection of the [N II] 122 μm and [O III] 52 μm lines". In: MNRAS 521.2 (May 2023), pp. 2526–2534. DOI: 10.1093/mnras/stad687. arXiv: 2211.01424 [astro-ph.GA].

[15] M. Kaasinen, ..., **M. Killi**, et al. "To see or not to see a $z \sim 13$ galaxy, that is the question. Targeting the [C II] 158 μm emission line of HD1 with ALMA". In: A&A 671, A29 (Mar. 2023), A29. DOI: 10.1051/0004-6361/202245093. arXiv: 2210.03754 [astro-ph.GA].

[16] C. M. Casey, A. Cooray, **M. Killi**, et al. "Near-infrared MOSFIRE Spectra of Dusty Star-forming Galaxies at $0.2 < z < 4$". In: ApJ 840.2, 101 (May 2017), p. 101. DOI: 10.3847/1538-4357/aa6cb1. arXiv: 1703.10168 [astro-ph.GA].